

Ratri Mukherjee

Ph.D. Researcher | Machine Learning | NLP | Biomedical AI

Iowa City, IA | ratri-mukherjee@uiowa.edu | (903) 521-8598

LinkedIn: ratri-mukherjee | GitHub: ratri1310 | Google Scholar: ratri-mukherjee

Research Profile

Ph.D. researcher in Electrical & Computer Engineering focused on machine learning, natural language processing, biomedical informatics, and knowledge-guided learning. Builds zero-shot and generalized zero-shot text classification methods evaluated on 1.5M+ PubMed abstracts and 29K+ MeSH concepts, using PyTorch, Hugging Face Transformers, contrastive learning, knowledge graphs, GNNs, RAG, and GPU-based model development. Peer-reviewed publications at EMNLP, IEEE ICDM, PAKDD, IEEE BIBM, and IEEE EMBC.

Technical Skills

Programming: Python, SQL, MATLAB
ML / AI: PyTorch, Hugging Face, Transformers, LLMs, Retrieval-Augmented Generation, Contrastive Learning, Graph Neural Networks, Zero-Shot Learning
Data / Infrastructure: Pandas, NumPy, MongoDB, Neo4j, Docker, Git, Linux, AWS, GCP
Domains: Natural Language Processing, Biomedical Informatics, Knowledge Graphs

Research & Professional Experience

University of Iowa

Graduate Research Assistant

Iowa City, IA

Aug 2023 – Present

Designed and evaluated Transformer-based deep learning frameworks for large-scale NLP and multi-label classification, conducting experiments over 1.5M+ documents and 29K+ labels.

Developed representation learning and model generalization methods combining pretrained language models, contrastive learning, and structured knowledge to address sparse, imbalanced, and limited-supervision learning settings.

Built end-to-end ML research and experimentation pipelines covering data processing, GPU model training and large-scale evaluation across multiple datasets and model architectures.

University of Texas at Tyler

Graduate Research Assistant

Tyler, TX

Aug 2021 – Jul 2023

Developed ML/DL pipelines in Python, PyTorch, and scikit-learn for an NIH-funded project predicting life-threatening apnea events in preterm infants from physiological time-series data.

Built preprocessing, feature-extraction, classification, and evaluation workflows for biomedical signal modeling.

Published and presented results at IEEE EMBC; contributed to a Pediatric Research manuscript under review.

Catalyst Solutions

Data Analyst

Dhaka, Bangladesh

Nov 2018 – Jul 2021

Built forecasting models on healthcare and finance datasets using SQL and Python; improved data-processing pipelines, increasing system efficiency by 20%.

Selected Projects

AI-Enabled Healthcare Assistant

Python, OpenAI API, Flask, RAG, TTS

Built a voice-enabled AI assistant for older adults and caregivers using retrieval-augmented generation and text-to-speech to provide real-time gerontological care information.

Selected Publications

SPUR-XMC: Context-Adaptive Spurious Correlation Mitigation for Extreme Multi-Label Classification of Biomedical Literature Empirical Methods in Natural Language Processing(EMNLP, 2026.

Disentangled Contrastive Representation Learning for Zero-Shot Biomedical Text Classification. IEEE International Conference on Data Mining (ICDM), 2025.

Semantic Knowledge Augmented Hypergraph Contrastive Learning for Zero-Shot Biomedical Text Classification. Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD), 2025.

Context-Aware Contrastive Representation Learning for Zero-Shot Biomedical Text Classification. IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 2024.

Education

University of Iowa

Ph.D. in Electrical & Computer Engineering | GPA: 3.91

Iowa City, IA

2023 – Present

University of Texas at Tyler

M.S. in Electrical Engineering | GPA: 4.00

Tyler, TX

2021 – 2023

Rajshahi University of Engineering & Technology (RUET)

B.S. in Electrical & Electronic Engineering

Bangladesh

2013 – 2017

Awards & Honors

ICDM Travel Award (2025)

Bob Herd Endowed Scholarship (2023)

Best Paper – ICEEE (2017)

PAKDD Travel Award (2025)

3MT Winner – UT Tyler (2023)

Graduate Fellowship – UT Tyler